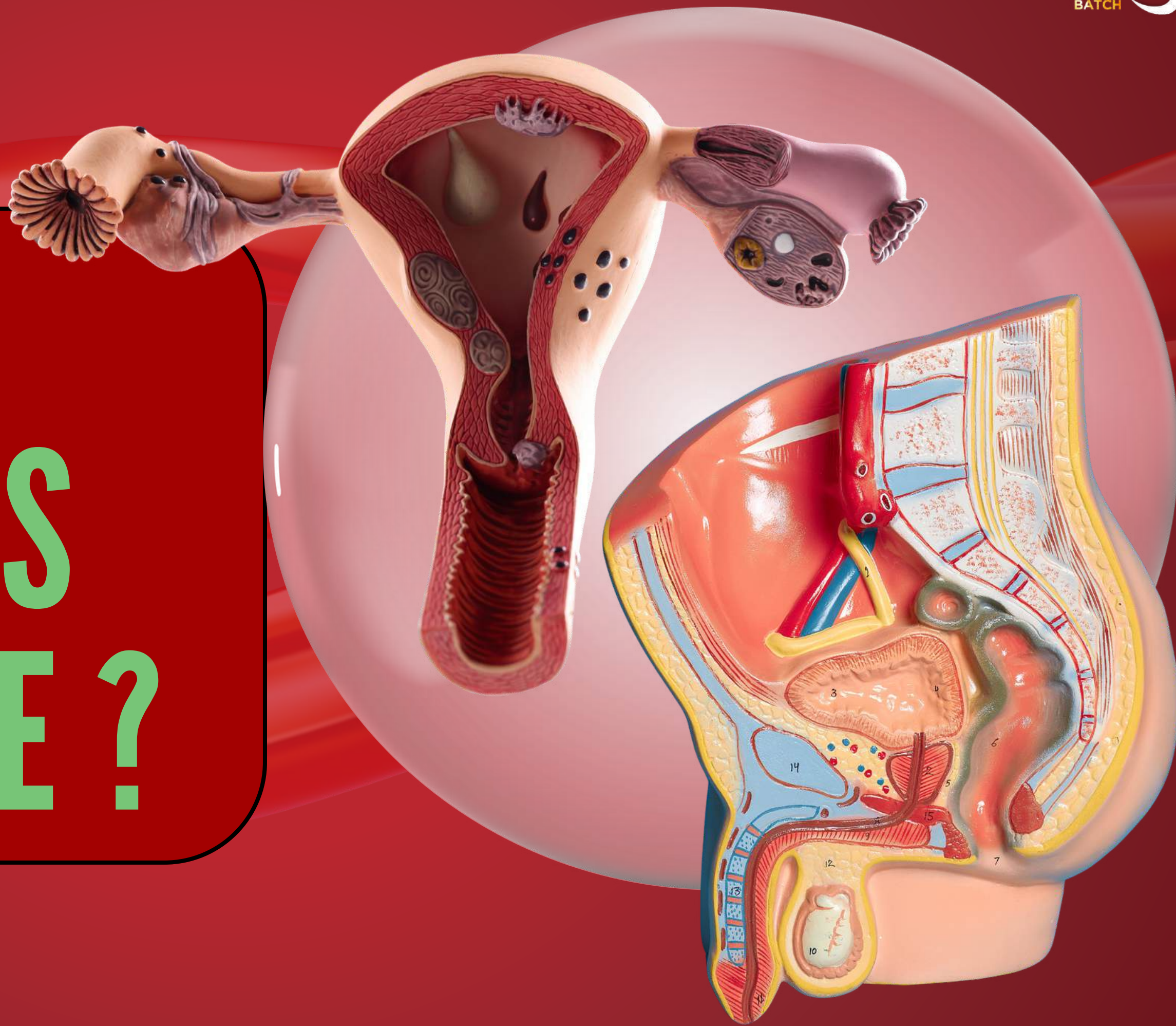


L5

CLASS X - SCIENCE

HOW DO ORGANISMS REPRODUCE ?

PRASHANT KIRAD



Female Reproductive System

Primary sex organs — Ovaries

- Produce eggs (female germ cells / ova).
- Also produce female sex hormones (oestrogen and progesterone).

Egg release

- At puberty, some eggs begin to mature.
- One egg is released every month by one ovary (ovulation).

Path of egg: Ovary → Oviduct / Fallopian tube → Uterus

Role of oviduct / fallopian tube

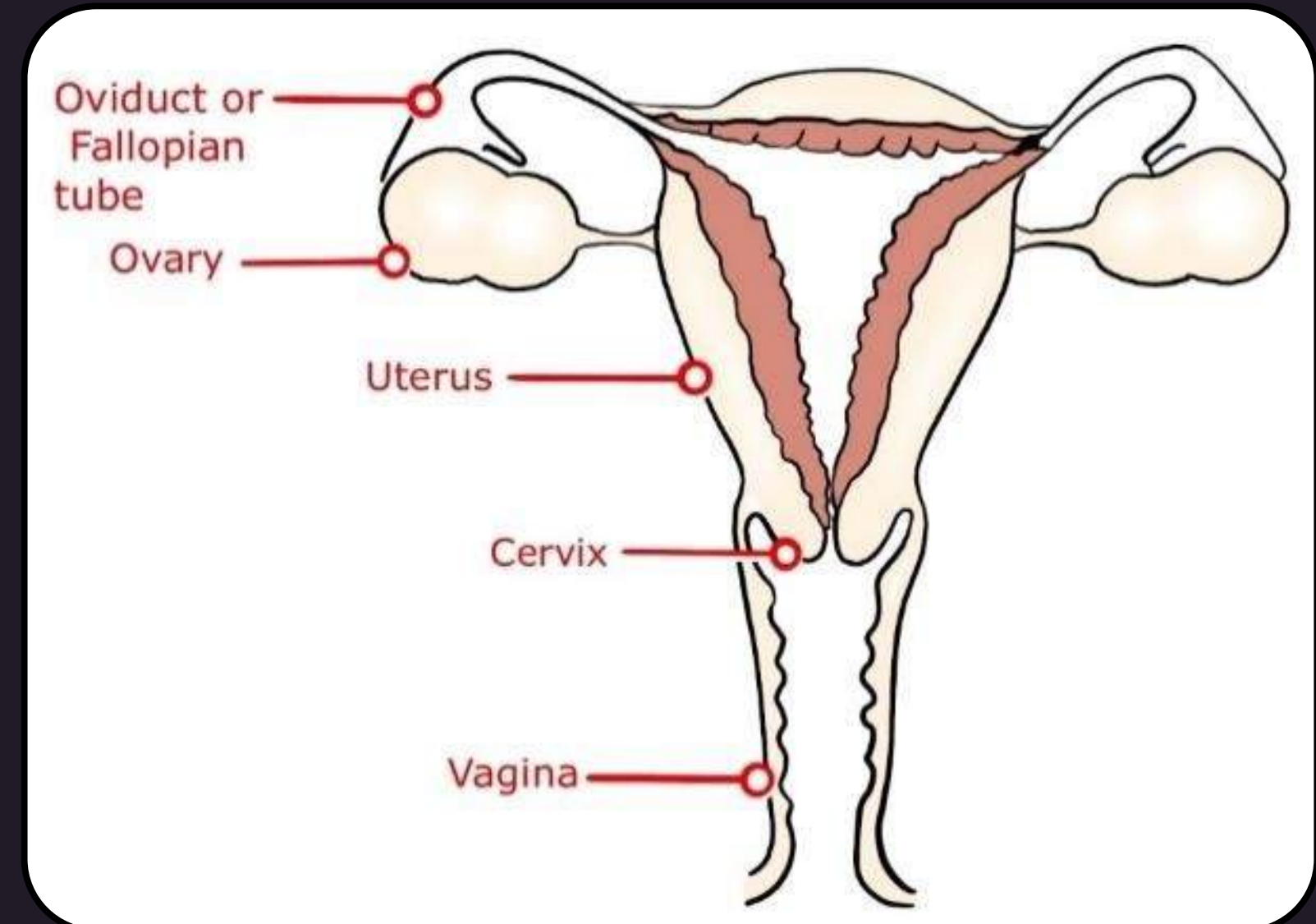
- Carries the egg from ovary to uterus.
- Fertilisation usually occurs here.

Role of uterus

- Muscular / elastic bag-like organ.
- Receives the fertilised egg.
- Embryo develops here.

Cervix and vagina

- Uterus opens into the vagina through the cervix.
- Sperms enter through the vaginal passage.



If fertilisation occurs:

- Zygote forms.
- Embryo gets implanted in the uterus.
- Embryo develops into a foetus.

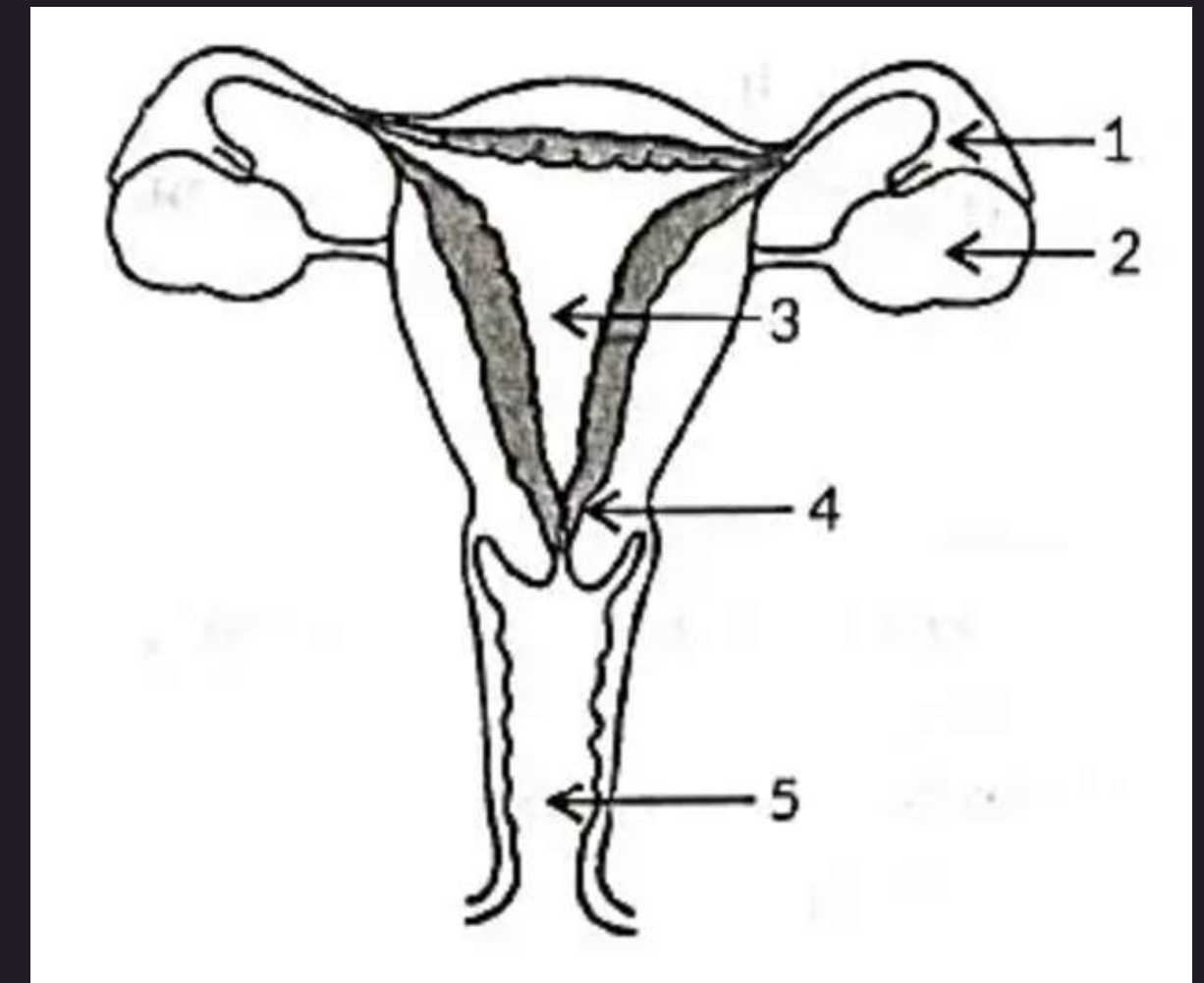
Organ	Location / Structure	Function
Ovary	Paired organs in the lower abdomen, one on each side of uterus.	Produce female gametes (ova/eggs) and secrete hormones (estrogen, progesterone) .
Ovulation	Process in the ovaries, once every month.	Release of a mature ovum from an ovary.
Fallopian Tubes (Oviducts)	Thin tubes extending from ovaries to uterus.	Carry egg from ovary to uterus; site of fertilisation .
Uterus (Womb)	Hollow, muscular organ in pelvic cavity.	Embryo implants in uterine lining (endometrium); supports development into foetus .
Cervix	Narrow, lower part of uterus.	Connects uterus to vagina; passage for sperms; dilates at childbirth .
Vagina	Muscular canal from cervix to outside of body.	Receives sperms during intercourse; acts as birth canal at delivery.

Check yourself

Q. Identify the given diagram. Name the parts 1 to 5

Ans: (a) The given diagram is the sectional view of the human female reproductive system. The labeled parts are:

1. Funnel of fallopian tube or oviduct
2. Ovary
3. Uterus or womb
4. Cervix
5. Vagina



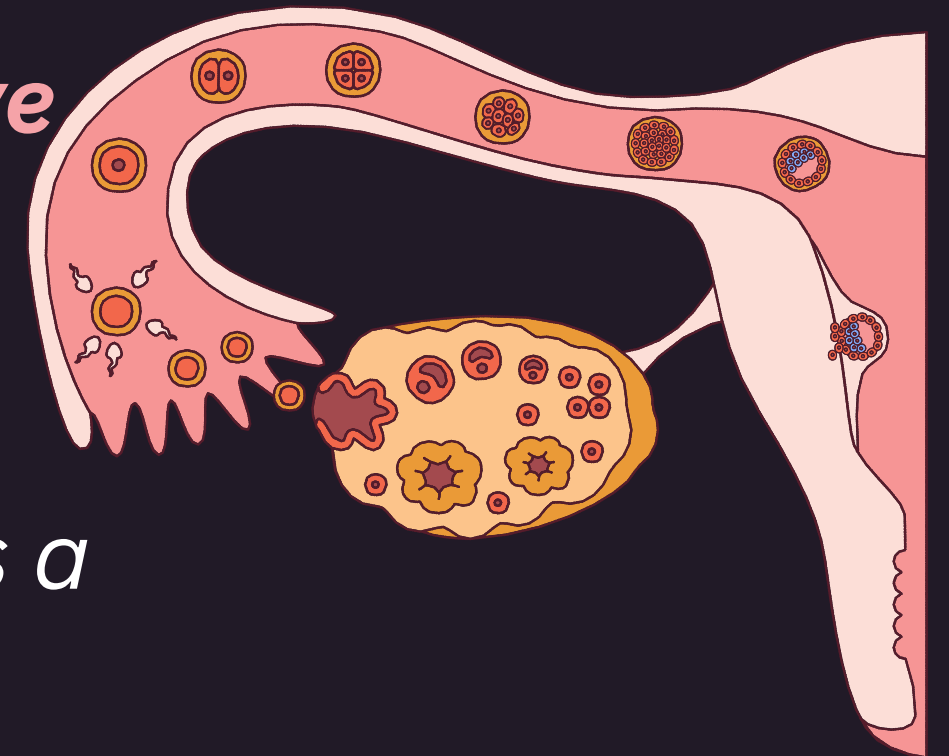
2017, 2018, 2020, 2021

Q. Write the functions of the following in the human female reproductive system: (a) Ovary (b) Oviduct (c) Uterus (d) Fallopian tube

Part	Function
(a) Ovary	Produces female gametes (ova/eggs) and secretes female hormones – oestrogen and progesterone.
(b) Oviduct / (d) Fallopian Tube	Receives the ovum released from the ovary; site of fertilisation of sperm and egg.
(c) Uterus	Holds and nourishes the developing embryo till birth; its lining thickens to support implantation.

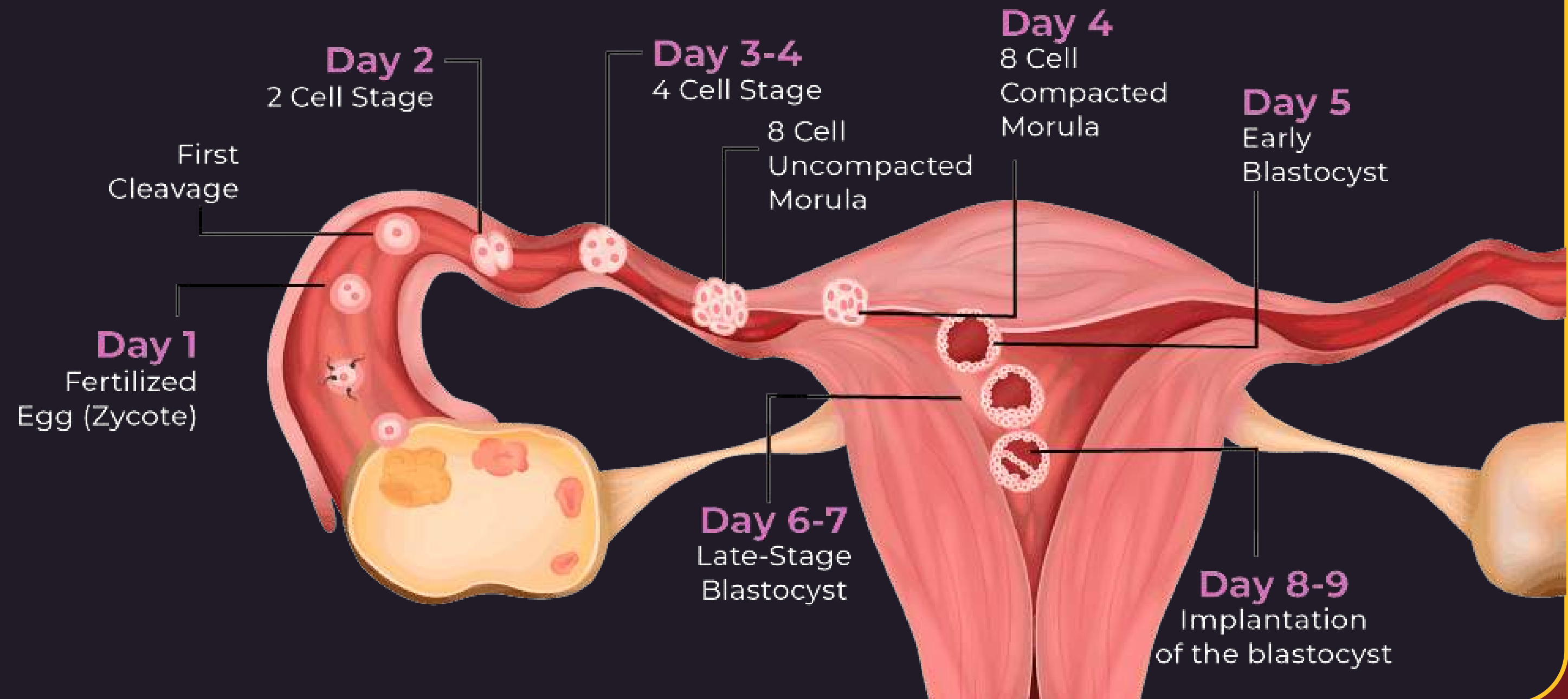
Process of Fertilization

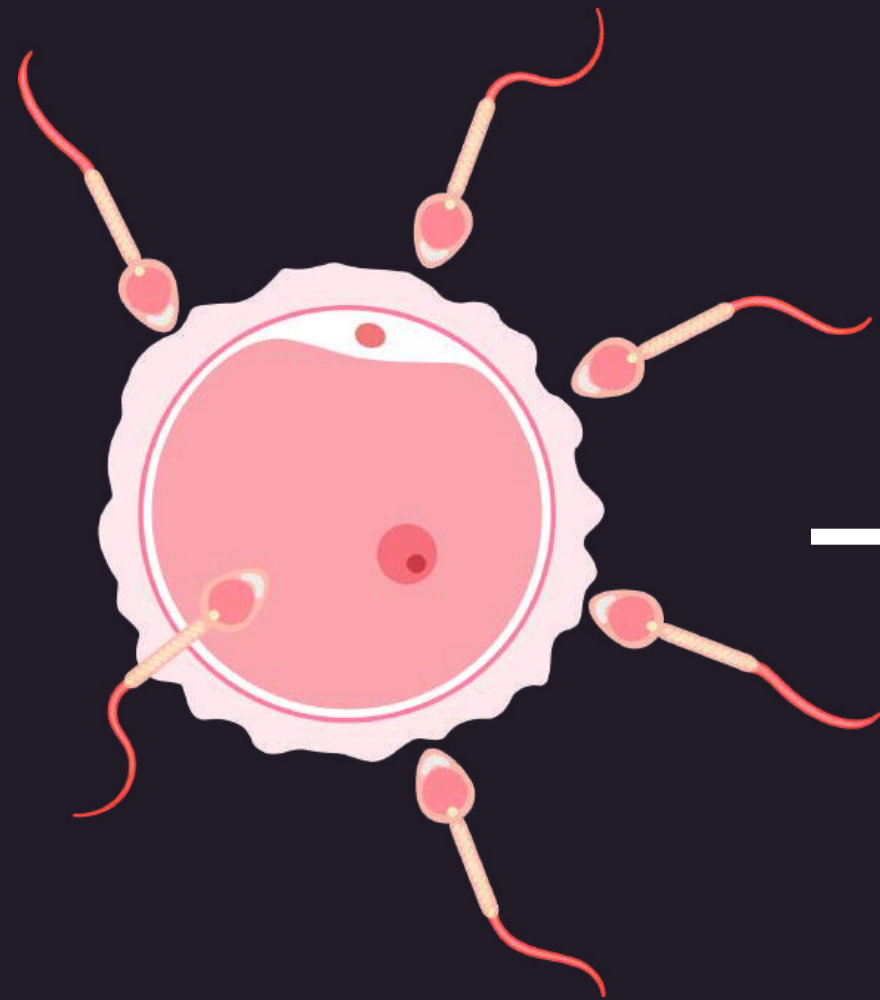
- During sexual intercourse, **millions of sperms** are released into the vagina.
- Sperms **swim upwards through the female reproductive tract using their tails** and reach the oviduct/fallopian tube.
- If an ovum is present, **one sperm fuses** with the egg in the oviduct. This fusion is called fertilisation and forms a zygote.
- The **zygote undergoes repeated cell divisions** to form an embryo while moving towards the uterus.
- The **embryo implants in the uterine lining** and gradually develops into a foetus.



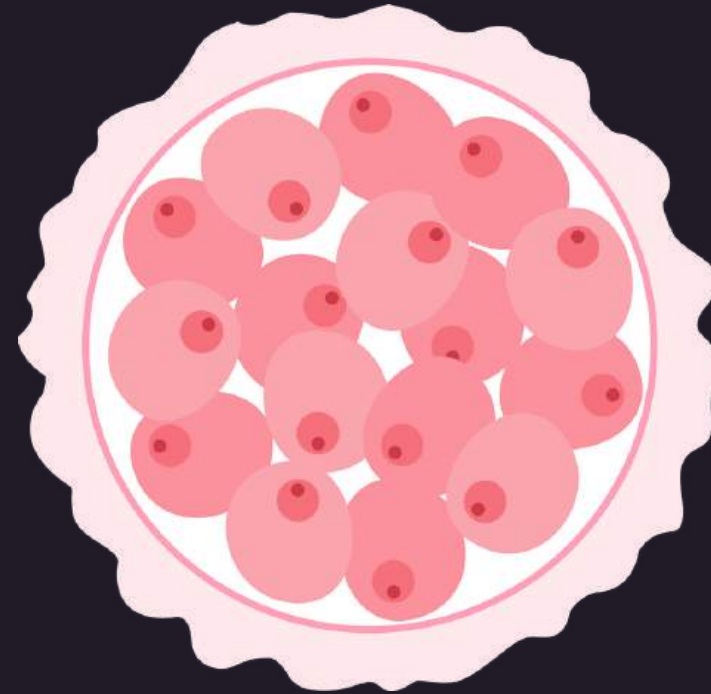
Implantation

- Implantation is the process where the fertilized egg (zygote) attaches to the lining of the uterus (endometrium) to develop into an embryo.





Zygote



Embryo



Foetus



Baby

Placenta

What is placenta?

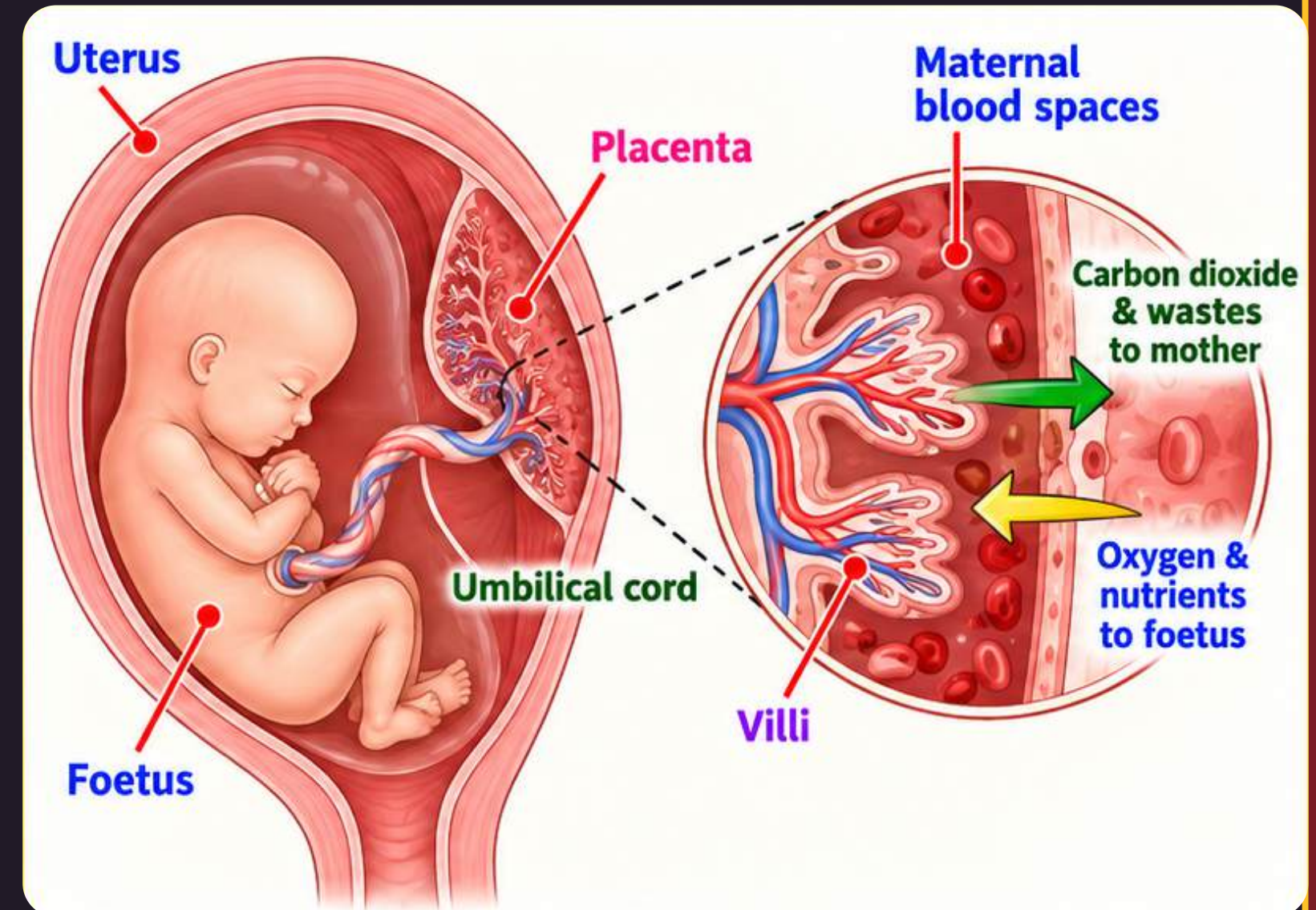
- Placenta is a **special tissue** that develops in the uterus during pregnancy.
- It connects the developing **embryo / foetus** with the mother.

Structure

- It is **disc-like** and attached to the uterine wall.
- It has **villi** on the embryo side.
- On the mother side, **blood spaces** surround the villi.
- The foetus is connected to placenta by the umbilical cord.

Functions of placenta

- Transfers **oxygen and nutrients** from mother to foetus.
- Removes **carbon dioxide and waste products** from foetus to mother.
- Helps in **exchange of materials** between mother and foetus.



Importance

- Nourishes the developing foetus.
- Supports growth and development inside the uterus.

How does the birth occur?

A foetus develops inside the uterus.



Development continues for approximately 9 months



Rhythmic contractions of uterine muscles



Birth



Check yourself

Q. Where does fertilisation occur in human females? Explain how the sperm reaches the egg.

Answer: Fertilisation in human females usually occurs in the oviduct/fallopian tube.

During sexual intercourse, sperms are released into the vagina. They swim upward through the female reproductive tract and reach the oviduct. If an egg is present there, one sperm fuses with the ovum, forming a zygote.

CBSE 2011, 2014, 2016, 2018, 2020, 2023

Q. What is placenta? State its functions in human female. How does placenta help in the nourishment of the embryo?

Placenta is a special tissue connection between the embryo (developing baby) and the wall of the uterus in the mother. It is formed by the embryonic and maternal tissues.

Functions of Placenta:

- Provides nourishment to the embryo.
- Removes waste materials from the embryo's blood.
- Allows exchange of gases – oxygen and carbon dioxide.
- Transports nutrients and oxygen from the mother's blood to the embryo.
- Transfers antibodies and hormones from mother to the foetus.

How Placenta Helps in Nourishment:

The mother's blood rich in oxygen and nutrients passes into the embryo's blood through the placenta by diffusion, while waste products from the embryo diffuse back into the mother's blood for removal.

2017, 2018, 2020, 2022, 2024, 2025

Q.Explain the events that take place once a sperm reaches the oviduct till it becomes a foetus

Once a sperm reaches the oviduct:

- Fertilisation: Sperm fuses with ovum to form a zygote.
- Zygote divides repeatedly to form a multicellular embryo.
- Embryo moves to uterus and implants in the uterine wall.
- Placenta forms to supply nutrients and oxygen.
- The embryo grows and develops into a foetus with all body parts formed.

Check yourself

Ques: Differentiate between fertilisation and implantation. Mention the site of each in the female reproductive system.

Fertilisation	Implantation
It is the fusion of the male gamete (sperm) and female gamete (ovum) to form a zygote .	It is the attachment of the embryo to the lining of the uterus .
It occurs in the oviduct/fallopian tube .	It occurs in the uterus .

Check yourself

Ques: A developing embryo is successfully implanted in the uterus. As developing continues, it requires oxygen and nutrition and also produces waste substances. **Answer the following:**

1. Name the structure responsible for this exchange.
2. Where is it located?
3. **EXPLAIN** how oxygen and glucose reach the embryo?
4. How is the waste from the embryo removed?

1. The structure responsible for this exchange is the placenta.
2. The placenta is embedded in the uterine wall and connects the developing embryo with the mother.
3. Oxygen and glucose pass from the mother's blood to the embryo through the placenta. The placenta provides a large surface area for this exchange.
4. Waste substances produced by the embryo pass through the placenta into the mother's blood, from where they are removed by the mother's body.

Check yourself

Q. (a) In the female reproductive system of human beings, state the functions of:

(i) Ovary (ii) Oviduct.

(b) Mention the changes which the uterus undergoes, when

(i) it has to receive a zygote. (ii) no fertilisation takes place.

Ans: (a) (i) The ovaries in female are primary sex organs (or female gonads) which perform the dual function - production of female gametes (eggs or ova) and secretion of female sex hormones (estrogen and progesterone).

(ii) Oviducts or fallopian tube are paired tubes originating near the ovaries of their respective sides and extend upto uterus. The terminal part of fallopian tube is funnel-shaped with finger-like projections called fimbriae lying near ovary. Fimbriae pick up the ovum released from ovary and push it into fallopian tube.

Fertilisation also takes place in the oviduct.

(b) (i) As the ovary releases one egg every month, the uterus also prepares itself, every month to receive fertilised egg by making its lining thick and spongy to nourish the zygote if fertilisation takes place. (ii) When the female gamete/egg is not fertilised, this lining is not needed any longer. So, the lining slowly breaks and comes out through vagina as blood and mucus through menstrual cycle that takes place every month.

Check yourself

Q: Compare the reproductive parts of flowers and humans and answer the questions below:

(a) Which part of the human female reproductive system has a similar function as the stigma in a flower? Give a reason to support your answer.

(b) Testes in the male reproductive system would correspond to which part of the male reproductive system in a flower? Justify.

(c) The style of a flower and the fallopian tube in humans correspond functionally with each other. Is this statement true? Justify your answer.

(a) The vagina corresponds to the stigma.

Reason: The stigma receives pollen grains during pollination, similarly the vagina receives sperms during sexual reproduction.

(b) The anther corresponds to the testes.

Justification: The testes produce male gametes (sperms) in humans, while the anther produces male gametes (pollen grains) in flowers.

(c) Yes, the statement is true.

Justification: The style provides a passage for pollen tubes to carry male gametes to the ovary in flowers, similarly the fallopian tube provides a passage for the ovum and is the site where fertilization occurs in humans.

Check yourself

Question: A fertilized egg has successfully formed. However, the embryo fails to attach the uterine lining.

- 1. Name the process by which the embryo normally attaches to the uterine lining.**
- 2. Why is the uterine lining prepared for this process?**
- 3. What would normally happen to the embryo after successful attachment?**

- 1. The process is called implantation.**
- 2. The uterine lining becomes thick, spongy and richly supplied with blood so that it can receive and nourish the developing embryo.**
- 3. After successful implantation, the embryo continues to grow and develop in the uterus and gradually becomes a foetus.**

Check yourself

Ques: Arrange the following event in the correct sequence and briefly explain the transition.

Embryo → Fertilization → Foetus → Implantation → zygote → birth

Correct sequence: Fertilisation → Zygote → Embryo → Implantation → Foetus → Birth

- **Fertilisation:** Sperm fuses with the ovum in the oviduct.
- **Zygote:** The fertilised egg formed after fusion of the gametes.
- **Embryo:** The zygote divides repeatedly and forms an embryo.
- **Implantation:** The embryo attaches to the lining of the uterus.
- **Foetus:** The embryo continues to grow and develops into a foetus.
- **Birth:** After about nine months of development, the baby is born due to rhythmic contractions of the uterine muscles.

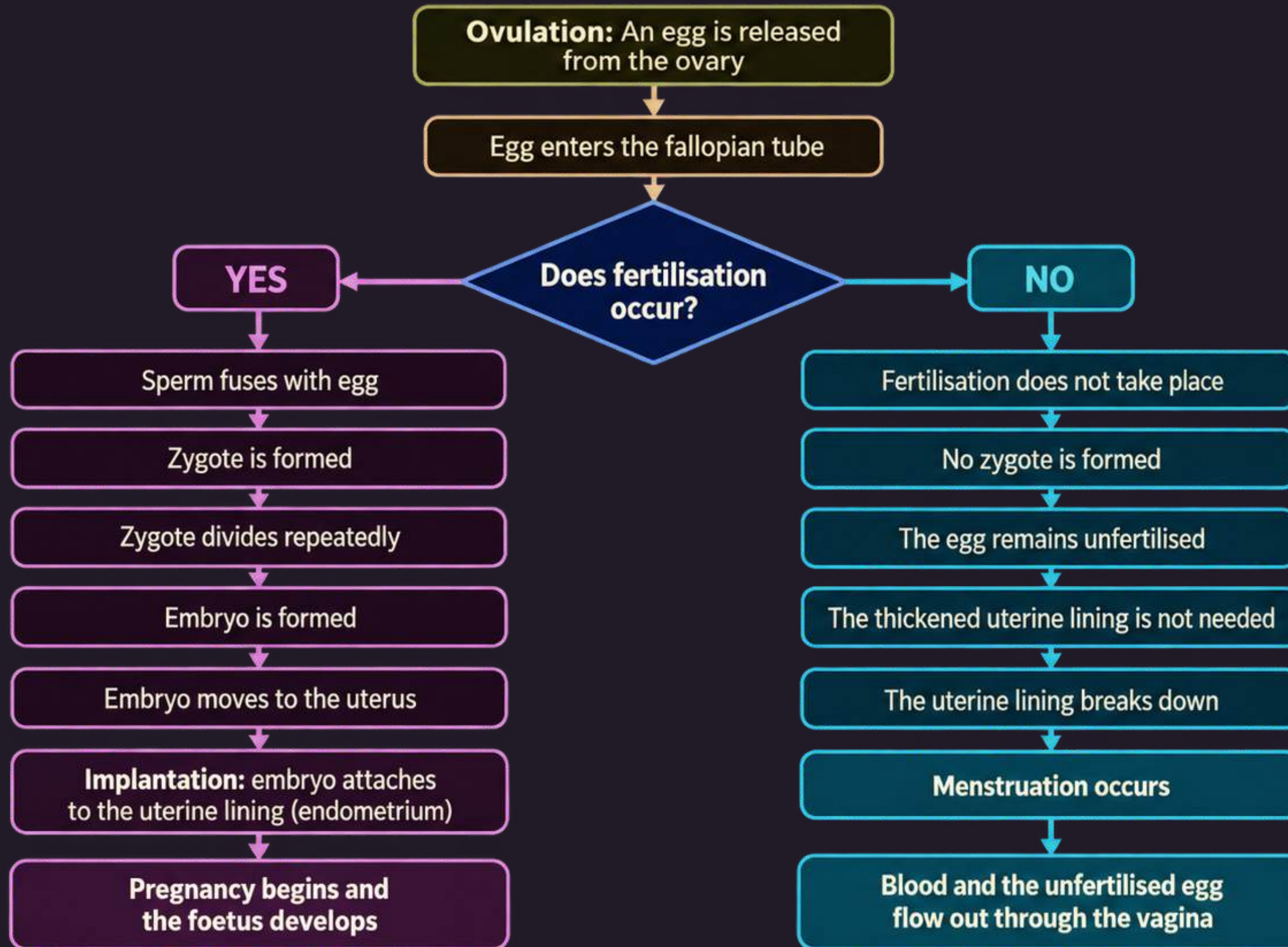
Check yourself

Question: A blockage develops in the fallopian tube of a woman. Her ovaries continue to produce eggs normally, but the egg cannot move normally through the blockage path.

1. Which structure normally carries the egg from the ovary towards the uterus?
2. In which structure does fertilization normally occur?
3. Explain why blockage of this path can interfere with fertilization/
4. Name the tissue that subsequently helps supply the developing embryo with oxygen and glucose?

1. The oviduct/fallopian tube normally carries the egg from the ovary towards the uterus.
2. Fertilisation normally occurs in the oviduct/fallopian tube.
3. If the fallopian tube is blocked, the egg and sperm may not be able to meet, so fertilisation can be prevented.
4. The tissue that later supplies the developing embryo with oxygen and glucose is the placenta.

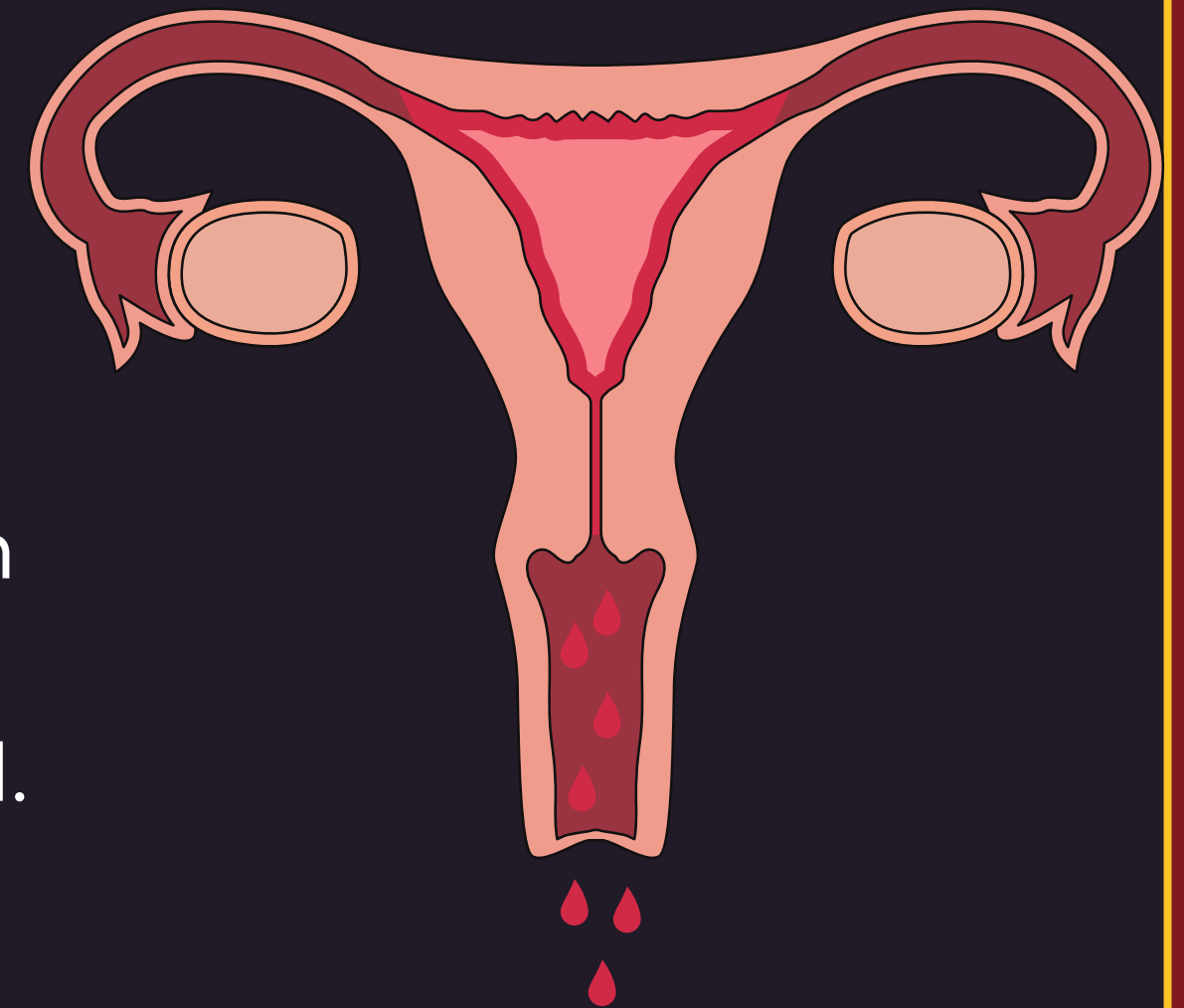
What Happens If Fertilisation Occurs / Does Not Occur?



MENSTRUATION.

↪ Occurs when fertilization does not occur

- The released egg **survives for about one day** if not fertilised.
- Since the ovary releases **one egg every month**, the uterus also prepares itself monthly to receive a fertilised egg.
- The **uterine lining becomes thick and spongy**, richly supplied with blood → to nourish the embryo if fertilisation had occurred.
- If fertilisation does not take place, this lining is not needed.
- The lining breaks down and comes out through the vagina as blood and mucus.
- This monthly cycle is called **Menstruation**.
- Duration: **Menstruation usually lasts for 2 to 8 days**.





Menarche and Menopause

Menarche

*Beginning of menstruation
At the time of puberty
when a girl gets
menstruation first
time.*



Menstruation

*The monthly shedding of the
uterine lining (blood and
tissues) when fertilisation does
not occur. It begins at
menarche and continues till
menopause, generally from 12
to 50 years of age.*



Menopause

*Stoppage of menstruation
At old age when a
women stops
menstruating.
usually occur around
the age of 45-55 years.*



Check yourself

**Q. (a) Why does the uterine lining become thick and richly supplied with blood?
(b) What would happen if the embryo failed to implant in the uterine lining?**

- (a) The uterine lining becomes thick, spongy and richly supplied with blood so that it can receive and nourish the developing embryo if fertilisation occurs.**
- (b) If the embryo fails to implant in the uterine lining, it will not continue normal development into a foetus because it cannot establish the required support and nourishment in the uterus.**

Same
You
Higher
Goals
😊

Kal Last Lecture Cover Karengenge

